

Full Featured IEEE Two-Column Paper Template

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Abstract—This template contains practical working examples for single/two-column equations, TikZ flowcharts, multi-column tables, matrix arrays, subfigure insertion, and automated BibTeX reference management inside an IEEE journal format.

Index Terms—IEEEtran, TikZ Flowchart, Subfigures, Equations, Matrix Arrays, BibTeX.

I. INTRODUCTION

THIS document demonstrates how to use graphics, arrays, flowcharts, and structured tables inside the IEEE LaTeX document class. Fundamental communication boundaries were mathematically established by Shannon [1]. In recent years, high-capacity neural network models [2] and attention-based architectures [3] have significantly advanced predictive modeling. Standard guidelines for formatting manuscripts follow official IEEE publishing standards [4].

II. EQUATIONS AND MATRIX ARRAYS

Standard aligned multi-line equation:

$$\begin{aligned} f(x) &= \int_0^\infty e^{-x^2} dx + \sum_{k=1}^n \frac{1}{k^2} \\ &= \frac{\sqrt{\pi}}{2} + \zeta(2). \end{aligned} \quad (1)$$

A 3×3 matrix array representation:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}. \quad (2)$$

As demonstrated in [1], matrix operations play a central role in channel capacity calculations.

III. TIKZ FLOWCHART CREATION

Fig. 1 shows a native TikZ block flowchart designed to fit cleanly inside a single column.

IV. IMAGE INSERTION

To insert standard figures or subfigures, use the setups below.

V. TABLE DESIGN

Table I demonstrates an IEEE-compliant table structure utilizing ‘booktabs’ styling, evaluated against state-of-the-art transformer baselines [3].

VI. CONCLUSION

This preamble provides all dependencies required for complex IEEE technical papers, with full citation support [1]–[4] linking directly to ‘references.bib’.

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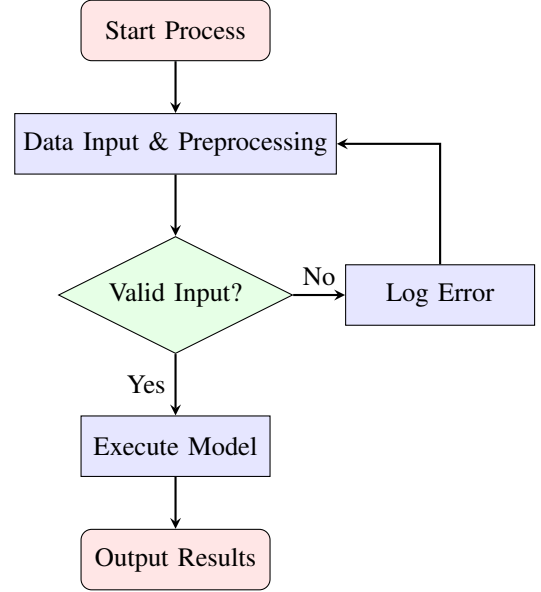


Fig. 1. System architecture and operational flowchart.

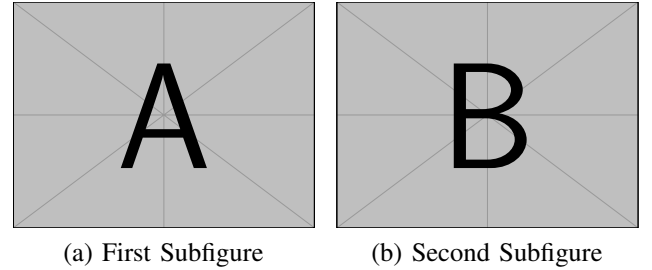


Fig. 2. Example of two subfigures side by side within one column.

TABLE I
PERFORMANCE COMPARISON ACROSS BASELINE MODELS

Method	Accuracy (%)	Latency (ms)	Parameters
Model A	88.4	12.5	1.2M
Model B	91.2	18.0	3.4M
Proposed	95.6	10.2	0.9M

REFERENCES

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